MILITARY SPECIFICATION

CHARGE, PROPELLING, XM195
PARTS FOR

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers the parts for XM195 Propelling Charges for 81MM mortar ammunition.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-2550	-Ammunition and Special Weapons, General Specification for

STANDARDS

MILITARY

MIL-STD-105	-Sampling Procedures and Tables for Inspection by Attributes (ABC-STD-105)
MIL-STD-109	-Quality Assurance Terms and Definitions
MIL-STD-1168	-Lot Numbering of Ammunition
MIL-STD-1169	-Packaging, Packing and Marking for Shipment of Inert Ammunition Components
MIL-STD-1235	-Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes

PRODUCT AND PACKING DRAWINGS

ARMY

9241321	-Container, Increment
9241322	-Cover, Container
9257039	-Charge, Propelling, XM195

FSC: 1315

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(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material.-Materials shall be in accordance with applicable drawings and specifications.

3.2 Parts.-The parts shall comply with all requirements specified on the parts drawings and with all requirements specified in the applicable specifications.

3.3 First Article Inspection.-This specification makes provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

3.4 Residue Test.-First article samples shall be tested in accordance with para. 4.4.1 to determine compliance with the Residue Test requirements specified therein.

3.5 Surface Defects.-Parts shall not exhibit blushing (as indicated by opaqueness), bubbles, obvious distortion, cracks, holes, tears or inclusions.

3.6 Workmanship.-All parts shall be fabricated and finished in a thorough workmanlike manner. They shall be free of dirt, grease and other foreign matter. All required markings shall be neat and sharply defined.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection.-Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to Standard MIL-STD-109 to define terms used herein. The provisions of Specification MIL-A-2550 shall apply.

4.1.1 Submission of Product.-At the time each completed lot of items deliverable under the contract is submitted to the Government for acceptance, the contractor shall supply the

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following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

a. A statement that the lot complies with all of the quality assurance provisions specified in this specification.
b. Drawing and specification number and date, together with an identification and date of changes thereto.
c. A statement that all material purchased by the contractor meets requirements, when such materail is controlled by Government or commercial specifications referenced in any of the contractual documents, and that certificates of conformance are on file and available for review.
d. Number of items in the lot.
e. Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.2 First Article Inspection

4.2.1 Submission.-The contractor shall submit a first article sample consisting of fifty (50) complete sets of parts (i.e. 50 each of every component and every subassembly) for evaluation in accordance with the provisions of 4.2.2. The sample shall be produced by the contractor or furnished by a supplier and shall have been manufactured using the same production processes, procedures and equipment which will be used in fulfilling the contract. All parts and materials, including packaging and packing, shall be obtained from the same source of supply as will be used in regular production. Prior to submission, the contractor shall inspect the sample to the degree necessary to assure that it conforms to the requirements of the contract and submit a record of this inspection with the sample, including statements of findings and certificates of conformance for materials (see 4.1.1c). A sample containing known defects will not be submitted unless specifically authorized by the contracting officer. A first article sample or portion thereof, as directed by the contracting officer, shall also be submitted whenever there is a lapse in production for a period in excess of 90 days or whenever a change occurs in manufacturing process, material used, drawing or specification such as to significantly affect product uniformity as determined by the Government.

4.2.2 Inspections to be performed.-Components will be subjected by the Government to any or all of the examina-

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tions or tests specified in 4.3 of this specification and any or all requirements of the applicable drawings.

4.2.3 Rejection.-If any assembly or component fails to comply with any of the applicable requirements, the first article sample shall be rejected. The Government reserves the right to terminate its inspection upon any failure of an assembly or component to comply with any of the stated requirements.

4.3 Quality Conformance Inspection

4.3.1 Lot Formation.-The term "inspection lot", as used in this specification, is defined as an essentially homogeneous collection of units of product from which a representative sample is drawn and inspected to determine conformance with applicable requirements. The sample selected shall represent only that quantity of units from which the sample was drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the inspection lot has been produced by one manufacturer, in one unchanged process, using the same materials and methods, in accordance with the same drawings, same drawing revisions, same specifications and same specification revisions, and complies with the provisions for submission of product as specified in Standard MIL-STD-105. Changes to the process, specifications or drawings, not affecting safety, performance, interchangeability or storage, as determined by the Government, shall not be deemed to alter the homogeneity of an inspection lot. All material submitted for inspection in accordance with this specification shall comply with the homogeneity criteria specified herein, regardless of the type of sampling procedure which is being applied to determine conformance with requirements. Lot numbering, as required, shall be in accordance with MIL-STD-1168.

4.3.2 Examination.-The examinations listed in the following subparagraphs shall be performed on inspection lots (see 4.3.1) of the components or assemblies designated in the applicable paragraph headings. Inspection for critical defects (and major defects, when so specified) shall be 100 percent. Sampling plans for major and minor defects shall be in accordance with MIL-STD-105, except that continuous sampling plans in accordance with MIL-STD-1235, may be used if approved by the Government. An AQL of 0.40 percent for each major defect and an AQL of 0.65 percent for each minor defect shall be applied to all of the major and minor defects listed in the applicable subparagraphs

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of this section, except where 100 percent inspection is specified. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.3.4.

4.3.2.1 Container, Increment (see dwg. 9241321).

Categories	Defects	Method Of Inspection	
Critical:	None defined		
Major:			
101.	Thickness of flange	Gage	(CD)
102.	Height of container	Gage	(CD)
103.	Inside diameter	Gage	(CD)
104.	Outside diameter without flange	Gage	(CD)
105.	Outside diameter with flange	Gage	(CD)
106.	Outside diameter angle, max	Gage	(CD)
107.	Inside diameter angle, max	Gage	(CD)
108.	Angle, max at leading end of container	Gage	(CD)
109.	Width of opening	Gage	(CD)
110.	Length of container without flange	Gage	(CD)
111.	Surface defects (see 3.5)	Visual	
Minor:			
201.	Workmanship (see 3.6)	Visual	

4.3.2.2 Cover, Container (see dwg. 9241322).

Categories	Defects	Method Of Inspection	
Critical:	None defined		
Major:			
101.	Diameter	Gage	(CD)
102.	Thickness	Gage	(CD)
103.	Surface defects (see 3.5)	Visual	
Minor:			
201.	Workmanship (see 3.6)	Visual	

4.3.3 Testing

4.3.3.1 Residue Test, Special Defect (see 6.3).-The contractor shall submit a first article sample of five hundred (500) sets of XM195 propelling charge component parts for loading, assembling and testing in accordance with the provisions of 4.4.1. The samples shall be selected from one lot of propelling charge component parts, and from the production process and using procedures and equipment which will be used in fulfilling the contract. The samples shall

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be inspected to assure that they conform to the applicable component drawings and to the inspection requirements of this specification. Defective samples will not be submitted.

4.3.4 Inspection equipment.-The inspection equipment required to perform the examinations and tests prescribed in this section is identified under the "Method of Inspection" headings in 4.3.2 (see 6.2).

4.4 Test Methods and Procedures

4.4.1 Residue Test.-This test shall be conducted at a Government proving ground. Test rounds (81MM- M374A3 mortar) shall be assembled with XM195 propelling charges. The propelling charges shall be loaded and assembled in accordance with Army drawing 9257039 (Charge, Propelling, XM195) using first article propelling charge component parts. Reference rounds shall be those selected from current production (81MM-M374A2 mortar rounds) at the time of first article testing.

Two hundred test rounds shall be test fired. These rounds shall consist of four firty (50) round groups. Each group shall be assembled at a different charge zone. The charge zones shall consist of charges one, two, three and four. Two hundred reference rounds shall be available for firing. These rounds shall consist of four fifty (50) round groups consisting of charge zones one, three, seven and nine.

The rounds shall be conditioned to and fired at 70 degrees Fahrenheit. All rounds shall be fired at an elevation of 800 mils, from an 81mm mortar tube (3.207 inches max.) for which the round is standard, at a rate of 12 rounds per minute. The mortar tube shall be cooled between the firing of each group. In addition, the tube shall be dry swabbed and elevation checked between the firing of each group. Observance shall be made for misfires.

The first article sample will be considered acceptable if no misfires occur. It will then be unnecessary to test the reference rounds. However, if a misfire occurs, it shall be determined if it was caused by residue from the propelling charge. Mortar tube elevation shall be checked. A misfire caused by residue will constitute rejection of the first article sample. In this event the reference rounds will be tested without delay, at the completion of the firing of that test round group, at that charge zone corresponding to that at which the test round misfired.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing and marking.-Pack in accordance with Code of Federal Regulations Title 49, Transportation Parts 100-199. Mark in accordance with MIL-STD-129. Marking shall comply with the Code of Federal Regulations.

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6. NOTES

6.1 Ordering data.-Procurement documents shall specify the following:

a. Title, number and date of this specification.
b. Data cards.-Data cards shall be prepared for each inspection lot in accordance with the information specified in Standard MIL-STD-1167.
c. Provisions for submission of first article samples.

6.2 Inspection Equipment Designs.-Inspection equipment designs are of two types-Government Special Inspection Equipment (SIE) designs and contractor designs. SIE designs are designated by drawing numbers under the "Method Of Inspection" headings in Section 4). Design responsibility for all other inspection equipment is assigned to the contractor. Equipment to be designed by the contractor is designated "CD". The contractor need not furnish any design when a complete Government SIE design is provided. Unless otherwise specified, however, the contractor may submit alternate or modified contractor designs of SIE in accordance with 6.2.2 and 6.2.3 should he elect to do so.

6.2.1 SIE designs.-SIE designs may consist of any of the following:

a. Detailed drawings which completely depict all information necessary for the fabrication and use of the item of inspection equipment.
b. A source control drawing or a specification control drawing as defined in MIL-STD-100.
c. An envelope drawing, as defined in MIL-STD-100, which establishes the criteria which a detail design shall meet. When envelope drawings are specified, the contractor shall prepare designs which comply with the criteria therein.

6.2.2 Contractor designs.-Contractor designs are required for all inspection equipment for which SIE designs are not specified and may include commercial equipment which the contractor proposes to use. (Commercial equipment is defined as unmodified equipment which is cataloged and available for purchase by the general public). Contractor designs shall include appropriate operating instructions, calibration procedures and maintenance procedures. Commercial equipment shall be fully described by catalog listings or other means which provide sufficient information to permit identification and evaluation by the Government and may include illustrations and engineering data. Designs shall be prepared for any special fixture(s) required to be used with commercial equipment, or with SIE designs if not

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otherwise covered thereby (see 6.2.1c). Designs shall be of the category and form (per MIL-D-1000) specified in the Contract Data Requirements Lists (DD Form 1423). The contractor is referred to MIL-HDBK-204, "Inspection Equipment Design" for guidance. The specification number and the applicable specific identifying information from Section 4 of this specification shall be referenced on each contractor design together with the component or assembly drawing number and revision letter to which the specific design applies.

6.2.3 Submission of designs for approval.-Contractor designs shall be approved by the Government prior to fabricating or procuring the equipment. Designs shall be submitted for approval in accordance with the stipulations, time frame and distribution specified in the Contract Data Requirements Lists (DD Form 1423) or in the contract. Partial submission of inspection equipment designs is permissible and encouraged. However, the completion date for design review will be based on the date of the final submission of designs. Picatinny Arsenal design review will normally be accomplished within one month after receipt.

6.3 Special defects

6.3.1 Definition.-A special defect is a defect that judgement and experience indicate could, depending on the degree of variance from the design requirement, result in hazardous or unsafe conditions for individuals using, maintaining or depending upon the product.

6.3.2 Disposition.-Upon investigation and analysis, of lots exhibiting special defects, by the responsible technical activity, a recommendation of acceptance or rejection shall be made to Director of Quality Assurance, U.S. Army Armament Command. Final acceptance of such lots on waiver shall be made only by the Director of Quality Assurance, U.S. Army Armament Command.

6.4 Submission of data.-One copy of proving ground test data (i.e. firing record containing Residue Test results) will be forwarded to each of the following:

Commander
Picatinny Arsenal
ATTN: SARPA-QA-A-A, SARPA-AD-E-A4
Dover, New Jersey 07801

Custodian:
ARMY-PA

Preparing Activity:
ARMY-PA

Project Number: 1315-A001

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